

1.1 Analytical Question: The Duality of PCA Objectives

Provide a formal proof demonstrating that for a centered dataset, the objective of maximizing the variance of the projected data is equivalent to the objective of minimizing the mean squared reconstruction error. Supplement your proof with a geometric argument, possibly using a diagram, that illustrates this relationship in a simple 2D or 3D case. Explain why this duality is significant for interpreting PCA's function.

2.1 Analytical Question: The Impact of Data Scaling

Discuss the sensitivity of Principal Component Analysis to the scaling of the input features. Explain why performing PCA on a raw, unstandardized dataset can lead to misleading principal components. Use a hypothetical example (e.g., one feature measured in kilometers and another in millimeters) to illustrate your point. Conclude by outlining the conditions under which you would choose *not* to standardize the data before applying PCA.

3.1 Analytical Question: The Centering Problem in Feature Space

In Kernel PCA, data is implicitly mapped to a high-dimensional feature space $\mathcal{F}$ via a map Φ . Standard PCA requires the data to be centered in the space where the analysis is performed. Explain why we cannot simply compute the mean $\mu_\Phi = \frac{1}{N} \sum_{i=1}^N \Phi(\mathbf{x}_i)$ and subtract it from each $\Phi(\mathbf{x}_i)$. Then, starting from the desired centered kernel matrix $\mathbf{K}_{\text{centered},ij} = \langle \Phi(\mathbf{x}_i) - \mu_\Phi, \Phi(\mathbf{x}_j) - \mu_\Phi \rangle$, derive the well-known expression for centering the kernel matrix $\mathbf{K}$ in terms of $\mathbf{K}$ itself and the $N \times N$ matrix of ones, $\mathbf{1}_N$.

4.1 Analytical Question: Critiquing PCA through the Lens of Algorithmic Fairness

Classical PCA's objective is to minimize the global average reconstruction error. Citing recent research directions, explain how this global objective can lead to unfair outcomes when the data contains distinct demographic groups. Define what "fairness" might mean in the context of dimensionality reduction, and describe the core idea behind one approach to "Fair PCA" (e.g., minimizing the maximum reconstruction loss across groups).

7.1 Analytical Question: Asymptotic Properties of MLE

Maximum Likelihood Estimators (MLEs) are widely used in machine learning. Beyond their intuitive appeal, they possess several desirable asymptotic properties. Define and explain the significance of the following three properties in the context of training and evaluating machine learning models:

1. Consistency
2. Asymptotic Normality
3. Efficiency (or Asymptotic Efficiency)

1.2 Numerical Problem: Manual PCA Calculation

Given the following 2D dataset $\mathbf{X} = \begin{pmatrix} 1 & 3 \\ 3 & 5 \\ 5 & 1 \end{pmatrix}$, perform the first step of Principal Component Analysis.

1. Calculate the mean and center the data.
2. Compute the 2x2 covariance matrix of the centered data.
3. Determine the first principal component (the eigenvector corresponding to the largest eigenvalue) of the covariance matrix.
4. Project the original data points onto this first principal component to obtain their 1D representations.

2.2 Numerical Problem: Eigendecomposition and Data Projection

A centered dataset has the following covariance matrix: $\mathbf{C} = \begin{pmatrix} 13 & -4 \\ -4 & 7 \end{pmatrix}$.

1. Calculate the eigenvalues of $\mathbf{C}$.
2. Find the corresponding orthonormal eigenvectors.
3. Determine the proportion of total variance explained by each principal component.
4. Given a new centered data point $\mathbf{x} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$, compute its transformed coordinates in the principal component space.

3.2 Numerical Problem: Kernel Matrix Computation and Centering

Consider the 1D dataset $\mathbf{X} = [-2, -1, 1, 2]$. You will apply Kernel PCA with a polynomial kernel $k(x, y) = (xy + 1)^2$.

1. Compute the 4x4 kernel (Gram) matrix $\mathbf{K}$.
2. Using the derived formula, compute the centered kernel matrix $\mathbf{K}_{\text{centered}}$.
3. Find the largest eigenvalue and corresponding eigenvector of $\mathbf{K}_{\text{centered}}$. (You may state the eigenvector up to a scaling factor).

4.2 Numerical Problem: Quantifying Representational Bias

A PCA model is used to reduce data from 10 dimensions to 2. The dataset contains two demographic groups, A and B. The total variance (trace of the covariance matrix) for data points in group A is 50, and the variance captured by the 2D projection for this group is 45. For group B, the total variance is 30, and the captured variance is 18.

1. Calculate the reconstruction error for each group.
2. Calculate the *percentage* of variance explained for each group.
3. Quantify the representational bias by computing the difference in the percentage of variance explained. Propose a simple modification to the PCA objective function using a weighted scheme to mitigate this bias.

7.2 Numerical Problem: Derivation of the MLE for the Laplace Distribution

The Laplace distribution, often used in robust regression, has the probability density function: $f(x|\mu, b) = \frac{1}{2b} \exp\left(-\frac{|x-\mu|}{b}\right)$. Given a set of i.i.d. observations $D = \{x_1, x_2, \dots, x_N\}$, derive the Maximum Likelihood Estimator for the location parameter μ , assuming the scale parameter b is known. (Hint: The derivative of $|x|$ is $\text{sgn}(x)$, and the objective function is not differentiable everywhere. Consider the properties of the function you are trying to maximize).